



2E10 Final Report (Part D)

Group X15

Throughout the buggy project, our team encountered several technical obstacles from both the bronze and silver challenges. While we successfully achieved the silver objectives, overcoming these challenges required significant time and effort, ultimately preventing us from attempting the Gold challenge within the allocated timeframe. This section details three major technical challenges faced, the reasons they proved difficult, and the steps taken to mitigate or resolve them.

1. Tuning the PID Controller for Silver Functionality.

For the Silver challenge we needed to implement the use of Proportional-Integral-Derivative (PID) control to manage buggy speed for two modes: maintaining a reference speed set via the GUI and smoothly following an object at a safe distance

Technical Detail: PID control requires careful tuning of three parameters – Kp (Proportional gain), Ki (Integral gain), and Kd (Derivative gain) – to achieve stable and responsive system behaviour without excessive overshoot or oscillation. This involved integrating feedback from sensors (wheel encoders for speed, ultrasonic sensor for distance) into the Arduino control loop to continuously adjust motor outputs based on the calculated error between the desired state (setpoint) and the actual state.

Why it was a Challenge: Our team relied primarily on a trial-and-error approach for tuning these PID parameters. This method proved to be extremely time consuming, involving numerous tests on the track, observing the buggy's behaviour (e.g., speed stability, smoothness of following, oscillations), adjusting the Kp, Ki, and Kd values in the code, and re-testing. Achieving optimal tuning was difficult, as changes to one parameter often needed adjustments to others, and sub-optimal tuning contributed to inconsistent behaviour during testing, complicating the debugging of other software components.

Resolution: Through persistent testing, we eventually arrived at a set of PID parameters that allowed the buggy to meet the functional requirements for the silver challenge demonstration. However, we recognise this tuning method was inefficient.

2. Ensuring Consistent Buggy Performance with Varying Battery Voltage

Maintaining predictable behaviour from the buggy proved difficult due to inconsistencies related to the power source.

Technical Detail: During development, we utilised two different power sources: the provided 7.2V rechargeable AA batteries and standard 9V AA batteries. We observed that the buggy's motors responded differently to the same Pulse Width Modulation (PWM) signals depending on which battery type was installed. The voltage supplied by either battery type decreased as it discharged, causing further variation in motor speed over time even with constant PWM signals.

Why it was a Challenge: This inconsistency was a significant hindrance. The PWM values calibrated to achieve specific speeds or behaviours (like moving straight), would often need re-adjustment when the battery type was switched or as the batteries discharged. This made systematic testing and debugging problematic, as it was difficult to isolate software issues from hardware-induced performance variations.

Resolution: We ultimately decided on using the 7.2V rechargeable batteries for the final test and demonstration. But we had to make sure these batteries were consistently charged to minimise voltage drop effects. However, the time spent messing around with the inconsistencies caused by switching between power sources in the project could have been saved by standardising the power source much sooner and may have enabled us with enough time to attempt the gold challenge.

3. Adapting the Steering Mechanism for Effective Line Following

Technical Detail: The initial approach considered was standard differential steering, where turns are achieved by driving both wheels forward but at different speeds. This method proved efficient for the bronze challenge. However it was not for silver, as during testing it became apparent that this method did not provide sufficiently sharp or responsive turning capabilities required for the specific track layout and line-following task. To address this, we implemented a pivot turning strategy. This involved driving one wheel forward while simultaneously driving the opposite wheel in reverse when a turn was indicated by the IR line-following sensors. For example, to turn right, the left motor would be driven forward and the right motor in reverse. Control during these turns was managed by commanding fixed, experimentally determined Pulse Width Modulation (PWM) values to each motor (e.g., setting one motor to a PWM value of approximately 165 and the reversed motor to 130), rather than relying on the PID controller which was used for regulating forward speed.

Why it was a Challenge: This change required a substantial redesign of the core motor control logic within the Arduino firmware. Translating the simple directional requirement ("turn left" or "turn right") from the IR sensors into the coordinated forward/reverse commands with appropriate PWM levels for each motor was complex. Determining suitable fixed PWM values for effective turning required considerable experimentation and tuning, especially since we had been implementing PID for the wheel speeds during turning prior to this. Ensuring smooth transitions between PID-controlled forward movement and this fixed-command pivot turning, and avoiding issues like wheel scuffing or overshooting the line, added to the difficulty.

Resolution: The pivot turning mechanism was successfully implemented and proved effective in enabling the buggy to navigate the required turns for line following. Through testing, specific PWM values were identified that provided a reasonable balance between turning speed and control. While potentially less smooth than ideally tuned differential steering for gentle curves, this pivot method provided the necessary agility for the track. The final implementation used this fixed-command pivot turn logic whenever the IR sensors indicated a deviation from the line, demonstrating a successful adaptation to overcome the limitations of the initial steering approach.